Sorted partial effects

Treatment effects across age and sex

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Session overview

1. Method and tutorial.
2. Simulated treated and control groups.
3. Sorted effects and characteristics of the tails.
4. Inference and interpretation.

Method and literature

01

Sorted partial effects describe how model-based effects vary across a population.

The original method and an applied R tutorial

The method and the tutorial

- **Original method:** sorted partial effects, classification analysis and inference (Chernozhukov, Fernández-Val, and Luo 2018).
- **R tutorial:** worked examples using `spe()` and `ca()` (Chen et al. 2020).
- The model supplies a partial effect for each covariate profile. Sorting summarizes the distribution of those effects.

What the analysis reports

Summary	Question
Average partial effect	What is the mean effect across the chosen population?
Sorted partial effects	How large are the effects at different percentiles?
Classification analysis	What characterizes profiles at the lower and upper tails?

The population and the effect scale determine all three summaries (Chen et al. 2020).

Simulated treatment comparison

02

A simple example with treated and control groups, age and sex.

A binary outcome and absolute risk differences

The example dataset

- **5,000 simulated participants**, randomized equally to `treated` and `control`.
- Outcome `Y = 1` is an adverse event over a common follow-up period.
- Age ranges from 40 to 80 years. Sex is binary in this teaching dataset.
- The simulated treatment effect varies with age and sex.

All values and coefficients are illustrative. The example represents no specific disease or treatment.

Benefit on the risk-difference scale

$$\Delta(x) = \Pr(Y = 1 \mid A = 0, X = x) - \Pr(Y = 1 \mid A = 1, X = x).$$

- `A = 0` is control. `A = 1` is treated.
- **Positive:** lower event risk under treatment. **Negative:** higher event risk.
- Multiplying the contrast by 100 gives **percentage points (pp)**.
- This is a conditional average contrast for a covariate profile.

Two profiles from the fitted model

Both profiles have `female = 0` and differ only in age.

Profile	Control risk	Treated risk	Benefit (pp)
Age 45	12.2%	19.6%	-7.3
Age 75	28.4%	8.5%	19.9

- Predict both treatment conditions for the same baseline profile.
- Subtract treated risk from control risk.
- Repeat this calculation for every profile in the dataset.

Model and sorted effects

03

The outcome model gives a risk difference for every observed age and sex profile.

Predictions, absolute benefits and their distribution

R: the outcome model

```r
fm <- outcome ~ group * age_z * female

fit <- glm(fm, data = trial, family = binomial())
```

- `age_z = (age - 60) / 10`.
- `*` includes main effects, pairwise interactions and the three-way interaction.
- The age pattern of the treatment odds ratio can differ by sex.

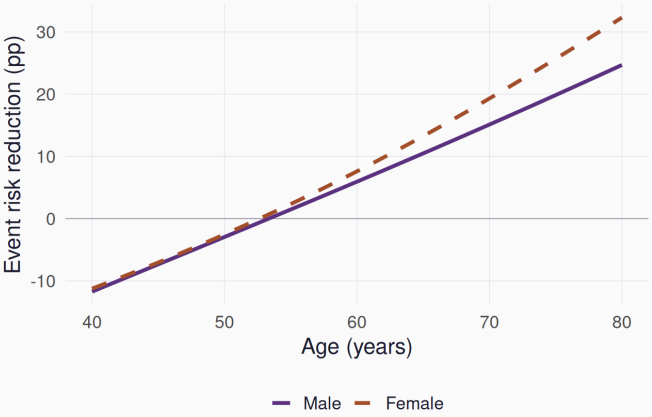
R: predictions under both groups

```r
control <- treated <- trial
control$group <- "control"
treated$group <- "treated"

benefit <- predict(fit, control, type = "response") -
  predict(fit, treated, type = "response")
```

- Keep each profile's age and sex fixed.
- Change only its treatment group.
- `type = "response"` returns probabilities.

Fitted benefit by age and sex



- The horizontal axis is **age**.
- Each line shows the fitted risk reduction for one sex.
- Values above zero favor treatment. Baseline risk and treatment interactions both shape the curves.

Simulated data. Benefit is control risk minus treated risk, in percentage points.

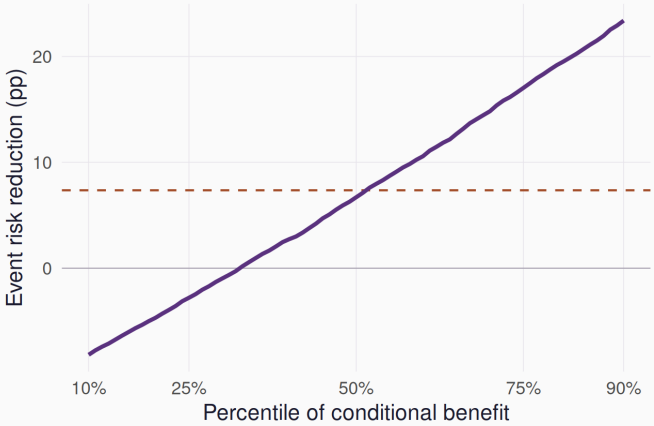
R: the mean and the sorted effects

```r
u <- seq(.10, .90, by = .01)
spe_curve <- Hmisc::wtd.quantile(
  benefit, weights = rep(1, nrow(trial)), probs = u
)
mean(benefit)
```

- The **mean** averages all 5,000 profile-specific benefits.
- The **SPE curve** reports quantiles of those same benefits.
- We display the central 10th–90th percentiles.

The weighted-quantile convention matches SortedEffects 1.7.0.

Sorted effects and the average



- The horizontal axis is the **percentile of estimated benefit**.
- The vertical axis shows risk reduction in **percentage points**.
- The dashed line is the mean: **7.4 pp**.

The curve increases because the effects have been sorted. Its shape alone is not a test of heterogeneity.

Interpreting the 80th percentile

The estimated 80th percentile is **19.2 pp**.

- Approximately 80% of the observed profiles have estimated benefits at or below this value.
- The remaining 20% have larger estimated benefits.
- This is a quantile of conditional effects. A quantile treatment effect compares outcome quantiles.

Characteristics of the effect tails

04

Classification analysis describes the profiles with the smallest and largest estimated benefits.

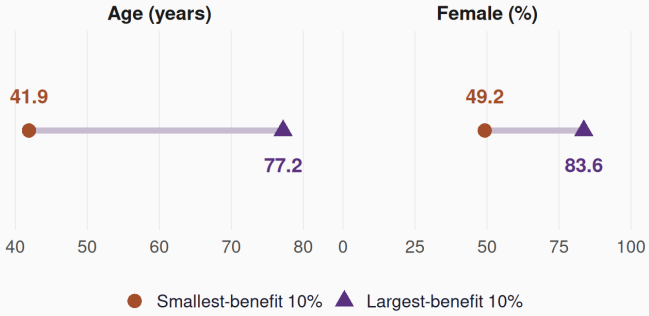
The lowest and highest 10%

R: selecting the benefit tails

```r
cutoff <- Hmisc::wtd.quantile(
  benefit, rep(1, nrow(trial)), c(.10, .90)
)
groups <- list(low = benefit <= cutoff[1],
               high = benefit >= cutoff[2])
lapply(groups, function(i)
  colMeans(trial[i, c("age", "female")]))
```

- Rank profiles by their estimated benefit.
- Select the lowest and highest 10%.
- Compare their mean age and proportion female.

Age and sex in the two tails



- Each dot is a **tail mean**: age or percentage female.
- The line connects the lowest- and highest-benefit groups.
- The panels use different units. The line length is descriptive.

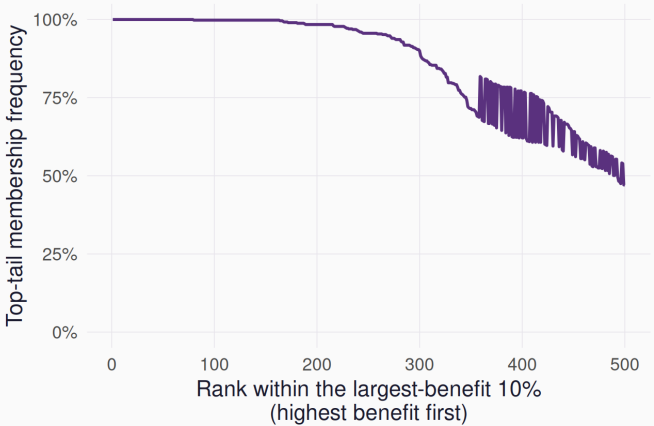
Tail groups are defined by estimated benefit. The connecting lines are not confidence intervals.

Tail comparisons and subgroup analyses

Analysis	Quantity compared
Treatment effect within an age/sex profile	Treated and control risks at the same covariates
Classification analysis	Mean characteristics in low- and high-benefit tails
Interaction analysis	Whether the treatment contrast varies with a covariate on the model’s scale

- A tail difference describes how the ranking relates to age or sex.
- It does not by itself test a treatment interaction.

Tail membership after refitting



- Start with the original **highest-benefit 10%**.
- Refit the model and reselect the tail in each bootstrap sample.
- The vertical axis is how often each original profile remains in that tail.

A separate sensitivity diagnostic. These frequencies are not individual membership probabilities.

Sampling uncertainty

05

Uncertainty includes model fitting and the resulting changes in the sorted effects and tail groups.

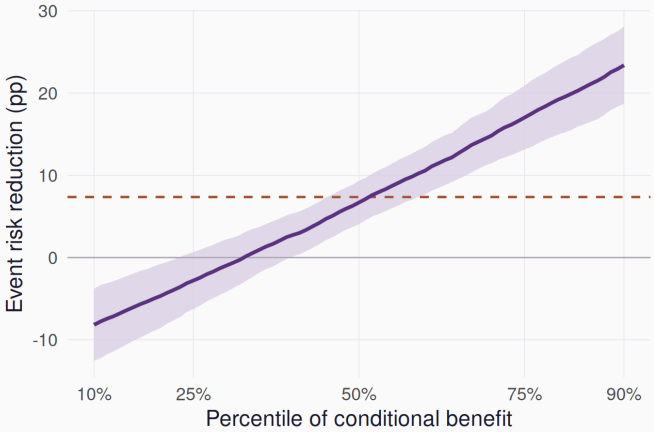
SPE confidence bands and classification comparisons

R: confidence bands for the SPE curve

```r
spe_fit <- SortedEffects::spe(
  fm = fm, data = trial, method = "logit",
  var_type = "categorical", var = "group",
  compare = c("treated", "control"),
  us = u,
  b = 499, alpha = .05, bc = FALSE, seed = 420
)
```

- `spe()` estimates the curve and its bootstrap confidence band.
- `compare` gives the **control-minus-treated** contrast.
- The example uses 499 repetitions and uncorrected point estimates.

Uncertainty across the sorted curve



- The solid line is the SPE curve; the dashed line is the average.
- Shading is a **95% uniform band** over the displayed percentile grid.
- Compare the band with zero when assessing the direction of effects.

SortedEffects 1.7.0, 499 bootstrap repetitions. The SPE calculation uses the observed profiles as its reference population.

R: uncertainty in classification

```r
ca_fit <- SortedEffects::ca(
  fm = fm, data = trial, method = "logit",
  var_type = "categorical", var = "group",
  compare = c("treated", "control"),
  u = .10, interest = "moment", cl = "diff",
  t = c("age", "female"),
  b = 499, alpha = .05, bc = FALSE, seed = 420
)
```

- Compare means in the highest- and lowest-benefit 10%.
- Refit, update the thresholds and reclassify in the bootstrap.
- Adjust the reported p-values jointly across **age and sex**.

Estimated differences between the tails

Characteristic	High minus low	Bootstrap SE	Joint p
Age (years)	35.29	1.04	0 / 499
Female (pp)	34.40	25.75	0.27

- **Positive differences:** higher mean age or proportion female in the high-benefit tail.
- The SE describes sampling uncertainty in the **difference**.
- Joint p-values account for these two comparisons.

Native `ca()` output, 499 repetitions. “0 / 499” means zero bootstrap exceedances, rather than an exact population p-value of zero.

Assumptions behind the estimates

- **Study design:** the treatment comparison has a defensible causal interpretation.
- **Outcome model:** the conditional risks and interaction structure are credible.
- **Reference population:** the covariate distribution matches the question.
- **Inference:** resampling reflects the design and the selected tail comparisons.

Interpretation and discussion

06

The analysis summarizes effect heterogeneity and describes where the fitted model places it.

The additional evidence needed for decisions

Treatment decisions

- The sorted curve shows how much the estimated benefit varies.
- Classification describes the baseline profiles at its extremes.
- Decisions also depend on harms, burden and the benefit threshold.
- A proposed treatment rule needs evaluation in independent data.

Questions for discussion

1. What does the sorted curve add to the average treatment effect?
2. When would age/sex plots or prespecified subgroup analyses be enough?
3. Which model choices would we vary in a sensitivity analysis?
4. What would we need before using an estimated ranking in practice?

Reading and resources

- **Tutorial first:** Chen et al. (2020), Sections 1, 2.2 and 2.3.
- **Original method:** Chernozhukov, Fernández-Val, and Luo (2018), introduction and Section 2.
- **Optional health application:** Pullabhotla and Souza (2022), Section 3, Figure 4 and Table 1.

The session page links to the papers, executable R example and inference notes.

Technical backup

+

Definitions, effect scales and resampling details.

Optional material for questions

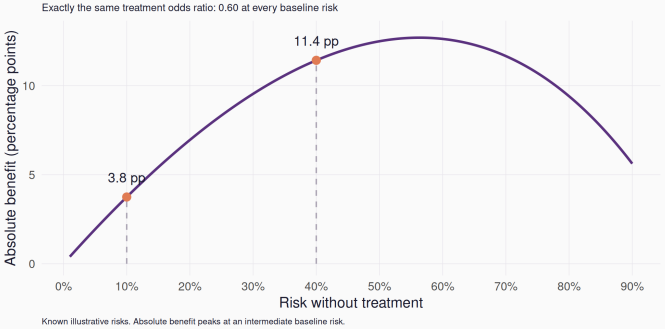
Definition of sorted partial effects

$$\mathsf{SPE}(u) = Q_{\Delta(X)}(u), \qquad \mathsf{APE} = E[\Delta(X)].$$

- The distribution of `X` defines the population.
- The average uses that full distribution.
- Classification describes the covariates in selected effect tails.

(Chernozhukov, Fernández-Val, and Luo 2018). Estimated curves use the package's weighted empirical quantile convention.

Effect heterogeneity depends on the scale



Analytic illustration, separate from the main age-by-sex interaction model.

- The odds ratio is **0.60 for every profile**.
- Absolute benefit varies with control risk.
- Risk-difference heterogeneity can occur without a treatment interaction on the log-odds scale.

Simulation assumptions

- Age is uniform from 40 to 80. The sex indicator is Bernoulli with probability 0.5.
- Control log odds include age, sex and their interaction.
- The treatment log odds ratio varies with age and sex.
- The fitted model matches the data-generating model.

The same 5,000 observed profiles define the primary SPE population. Coefficients and seeds are in `R/demo.R`.

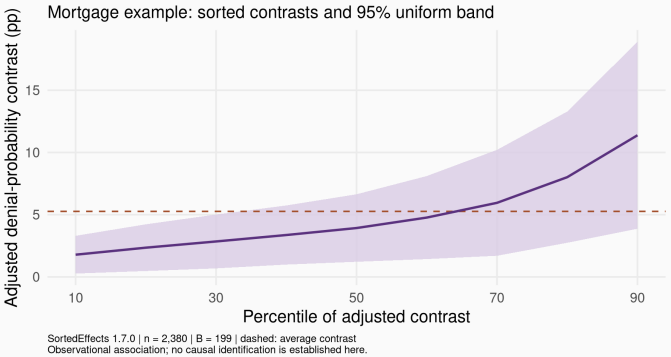
Resampling targets

- `spe()`**:** refit using bootstrap weights, then compute quantiles with the original profile weights.
- `ca()`**:** refit, update tail thresholds and recompute characteristic means with resampled weights.
- **Membership diagnostic:** refit and track the original profiles' selected-tail membership.

Classification output

- `ca(..., interest = "moment", cl = "diff")` reports high-minus-low mean differences.
- Bootstrap SEs quantify uncertainty in those differences.
- Joint p-values adjust across the supplied characteristics.
- The main table uses the native SEs and joint p-values.

The tutorial’s mortgage example



- Outcome: mortgage denial.
- Contrast: predicted denial risk at `black = 1` minus `black = 0`, with other covariates fixed.
- The curve summarizes an adjusted observational disparity.

Tutorial data (Munnell et al. 1996; Chen et al. 2020). Separate benchmark with uncorrected estimates and 95% bands.

References

Chen, Shuowen, Victor Chernozhukov, Iván Fernández-Val, and Ye Luo, "SortedEffects: Sorted causal effects in R," *The R Journal*, 12 (2020), 131–146.

Chernozhukov, Victor, Iván Fernández-Val, and Ye Luo, "The sorted effects method: Discovering heterogeneous effects beyond their averages," *Econometrica*, 86 (2018), 1911–1938.

Munnell, Alicia H., Geoffrey M. B. Tootell, Lynn E. Browne, and James McEneaney, "Mortgage lending in Boston: Interpreting HMDA data," *The American Economic Review*, 86 (1996), 25–53.

Pullabhotla, Hemant K., and Mateus Souza, "Air pollution from agricultural fires increases hypertension risk," *Journal of Environmental Economics and Management*, 115 (2022), 102723.